

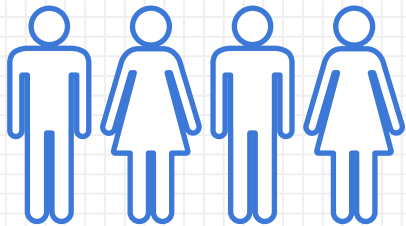


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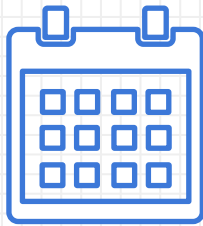
A Focused First Impression

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IDWSDS 2026



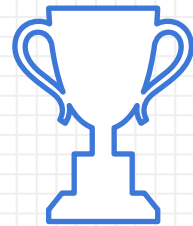


The course was offered for incoming 9th/10th and 11th/12th grade students.



Timeline

The course ran for one week over the summer with five six-hour days.



Goal

The course was designed to cover the broad (bio)statistical landscape and basic coding skills in R.

Challenges

Audience

- Math and data background
- Willingness to engage with course
- Application interests



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- Long days
- Accelerated schedule
- Dedicated project time
- Little to no homework

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- Balance between broad scope and enough detail
- Balance between foundations and application



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2. Data summarization, wrangling, and visualization in R
3. Regression mechanics, interpretation, and diagnostics
4. Navigating the scientific literature, presentation design, and giving critical feedback
5. Public speaking and statistics in the media



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- All coding topics were introduced with **sample code** and **practice questions** that could be answered by **tweaking** the sample code.
- Groups for each topic's activity were **randomized**.



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- All coding topics were introduced with **sample code** and **practice questions** that could be answered by **tweaking** the sample code.
- Groups for each topic's activity were **randomized**.
- Days 2-5 began with a **friendly competition** to recap past topics.



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Project

- Students were introduced to **six datasets** on Day 1, and they **ranked them** by interest level to help the instructor make groups.
- In **groups** of three or four, students designed **research questions** about their data and answered them with **descriptive statistics**, **visualizations**, and **regression**.



Project

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- In **groups** of three or four, students designed **research questions** about their data and answered them with **descriptive statistics**, **visualizations**, and **regression**.
- Each team **combined** their questions into a **presentation** that introduced the data and their **findings**.



[illegible]

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Overall Learning Goals

- I hoped students would leave this course with the instinct to **explore and question** real data they encounter.
- Both the **hands-on activities** and the **project** kept this goal in mind.

Disease risk in a state, like TN!

Measurement	Good	Bad (or possibly bad)
Counts of confirmed cases in each county at the end of the month (numerical, discrete)	- Helps policymakers decide how to send out resources - Consistent timing, can help you track trends over time - You have a good baseline for the lowest number	- Counties with big populations will have more cases - We miss people that are sick but haven't been diagnosed
Percent of folks over 18 that report having the disease in 2018 in each county (numerical, continuous)	- Helps you compare counties more fairly - Consistent timing - You have a good baseline for the lowest	- We rely on folks telling the truth - We set an arbitrary time frame, it could be thrown off by seasonality or another external event

Bias	Example	Consequence
Selection		
Non-response		
Voluntary		
Convenience		

- This policy had **mixed results**.

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- Students may come up with research questions that require **advanced data wrangling**
- Tech installation challenges may be **different on each machine**, causing some students to speed through setup and some to fall behind the group
- Choose the **level of depth** for your treatment of model diagnostics wisely
- Make sure you have **examples** ready to go for any activities where you ask students to **look for resources**



Students Really Enjoyed

- Working through code **examples**
- The group **project**
- Learning from **peers**
- **Recapping** each topic
- Support from the **teaching team** during activity time
- Having **access** to all course materials



Takeaways

- Giving **open-ended instructions** but having examples in your back pocket gives new statistics students the **flexibility** to explore the things that interest them most



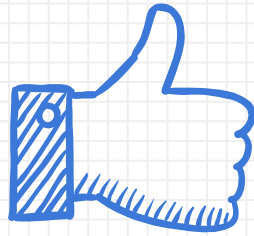
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Takeaways

- Giving **open-ended instructions** but having examples in your back pocket gives new statistics students the **flexibility** to explore the things that interest them most
- Providing space for students to **apply definitions** and abstract concepts to **their own interests** sparks thoughtful discussion
- Starting with the data landscape before jumping straight into coding builds **student confidence**



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THANKS!

Any questions?



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